

1. General Description

The EM74LVC541A is an 8-bit buffer/line driver with 3-state outputs. The device features two output enables ($\overline{OE}1$ and $\overline{OE}2$). A HIGH on $\overline{OE}n$ causes the associated outputs to assume a high-impedance OFF-state. Inputs can be driven from either 3.3 V or 5 V devices. This feature allows the use of these devices as translators in mixed 3.3 V and 5 V environments.

This device is fully specified for partial power down applications using I_{OFF} . The I_{OFF} circuitry disables the output, preventing the potentially damaging backflow current through the device when it is powered down.

2. Features and Benefits

- Wide supply voltage range from 1.2 V to 5.5 V
- Direct interface with TTL levels
- CMOS low power dissipation
- Overvoltage tolerant inputs to 5.5 V
- I_{OFF} circuitry provides partial Power-down mode operation
- Latch-up performance exceeds 250 mA
- Complies with JEDEC standards:
 - JESD8-7A (1.65 V to 1.95 V)
 - JESD8-5A (2.3 V to 2.7 V)
 - JESD8-C (2.7 V to 3.6 V)
 - JESD36 (4.5 V to 5.5 V)
- ESD protection:
 - HBM ANSI/ESDA/JEDEC JS-001 Class 3A exceeds 4000 V
 - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V

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3. Ordering Information

Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Quantity
EM74LVC541AD	LVC541A□ XYYWW	SOP-20L	plastic small outline package; 20 leads; body width 7.5 mm	2000
EM74LVC541APW	LVC541A□ XYYWW	TSSOP-20L	plastic thin shrink small outline package; 20 leads; body width 4.4 mm	3000

MARKING INFORMATION

NOTE: LVC541A□ = Serial number and Internal code; XYYWW = Vendor Code and Date Code.

LVC541A □

Internal Code

Serial number

X YY WW

Date Code -Week

Date Code -Year

Vendor Code

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4.Function Diagram

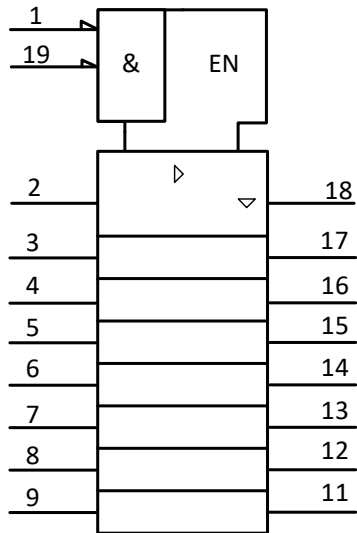


Fig. 1. IEC logic symbol

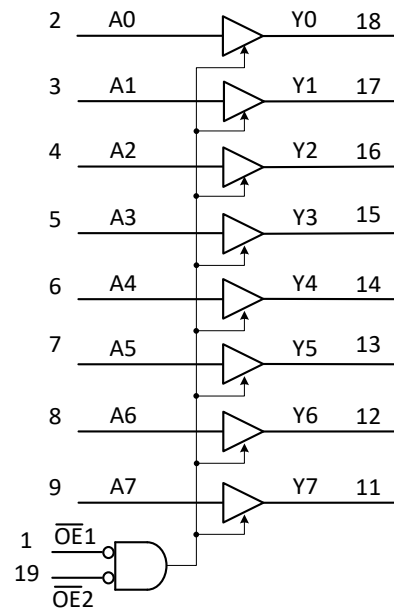


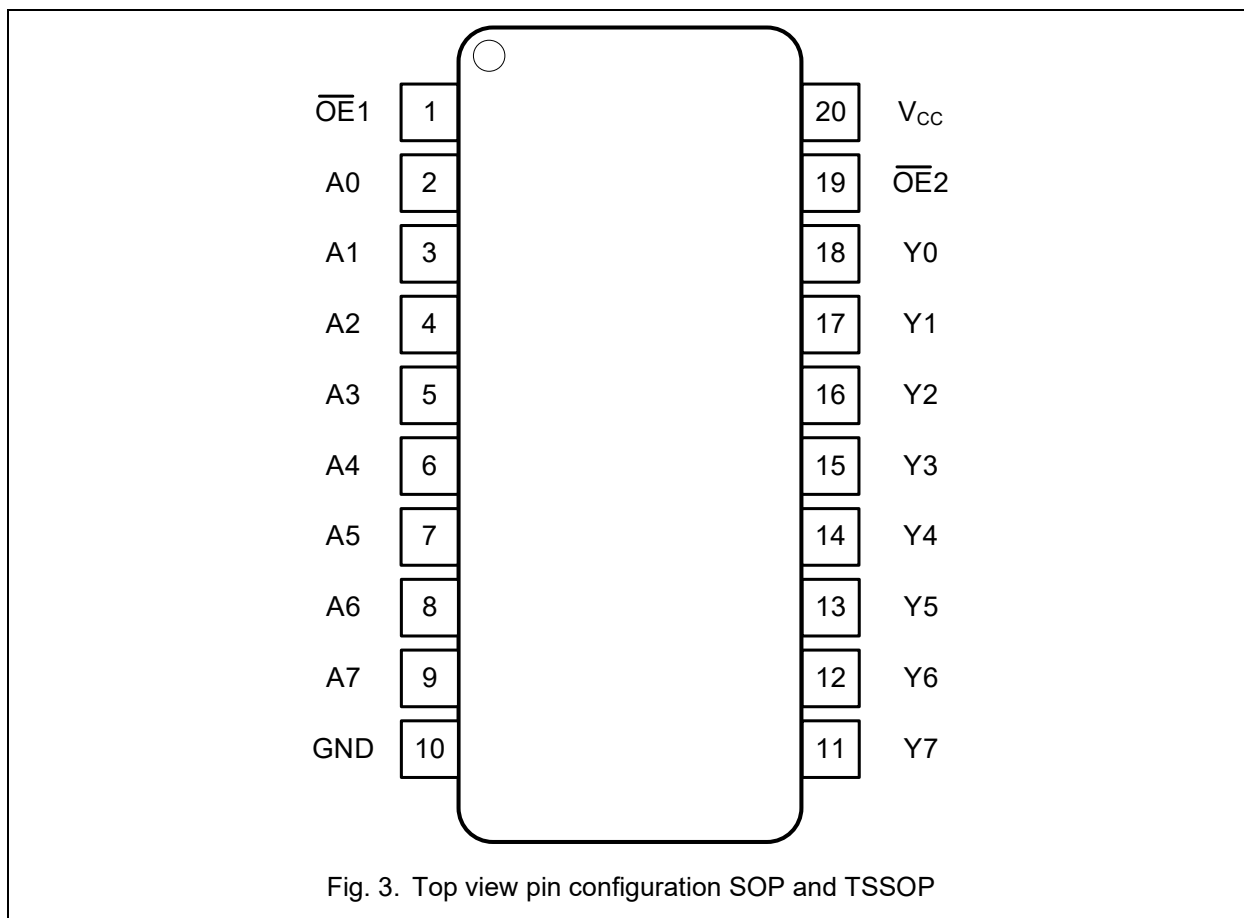
Fig. 2. Functional diagram

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5. Pinning Information

5.1. Pinning



5.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
$\overline{OE}1$	1	Output enable input (active LOW)
A0 to A7	2, 3, 4, 5, 6, 7, 8, 9	Data input
GND	10	Ground (0 V)
Y0 to Y7	18, 17, 16, 15, 14, 13, 12, 11	Data output
$\overline{OE}2$	19	Output enable input (active LOW)
V _{CC}	20	Supply voltage

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6.Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level; X = don't care; Z = high-impedance OFF-state.

Input			Output
$\overline{OE}1$	$\overline{OE}2$	An	Yn
L	L	L	L
L	L	H	H
X	H	X	Z
H	X	X	Z

7.Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V _{CC}	supply voltage		-0.5	6.5	V
I _{IK}	input clamping current	V _I < 0 V	-50		mA
V _I	Input voltage	[1]	-0.5	6.5	V
I _{OK}	output clamping current	V _O < 0 V or V _O > V _{CC}		±50	mA
V _O	output voltage	output HIGH or LOW [2]	-0.5	V _{CC} +0.5	V
		output 3-state	-0.5	6.5	V
I _O	output current	V _O = 0 V to V _{CC}		±50	mA
I _{CC}	supply current			100	mA
I _{GND}	ground current		-100		mA
P _{tot}	total power dissipation			500	mW
T _{stg}	storage temperature		-65	150	°C

[1] The minimum input voltage ratings may be exceeded if the input current ratings are observed.

[2] The output voltage ratings may be exceeded if the output current ratings are observed.

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8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V _{CC}	supply voltage		1.65		5.5	V
		functional	1.2			V
V _I	input voltage		0		5.5	V
V _O	output voltage	output HIGH or LOW	0		V _{CC}	V
		output 3-state	0		5.5	V
T _{amb}	ambient temperature		-40	25	125	°C
$\Delta t/\Delta V$	input transition rise and fall rate	V _{CC} = 1.2 V to 2.7 V			20	ns/V
		V _{CC} = 2.7 V to 3.6 V			10	ns/V
		V _{CC} = 4.5 V to 5.5 V			5	ns/V

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9.Static Characteristics

Table 6. Static characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V). Typical values measured at $T_{amb} = 25^{\circ}\text{C}$ (unless otherwise noted).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
V_{IH}	HIGH-level input voltage	$V_{CC} = 1.2\text{ V}$	1.08			1.08		V
		$V_{CC} = 1.65\text{ V to }1.95\text{ V}$	$0.65V_{CC}$			$0.65V_{CC}$		V
		$V_{CC} = 2.3\text{ V to }2.7\text{ V}$	1.7			1.7		V
		$V_{CC} = 2.7\text{ V to }3.6\text{ V}$	2.0			2.0		V
		$V_{CC} = 4.5\text{ V to }5.5\text{ V}$	$0.7V_{CC}$			$0.7V_{CC}$		V
V_{IL}	LOW-level input voltage	$V_{CC} = 1.2\text{ V}$			0.12		0.12	V
		$V_{CC} = 1.65\text{ V to }1.95\text{ V}$			$0.35V_{CC}$		$0.35V_{CC}$	V
		$V_{CC} = 2.3\text{ V to }2.7\text{ V}$			0.7		0.7	V
		$V_{CC} = 2.7\text{ V to }3.6\text{ V}$			0.8		0.8	V
		$V_{CC} = 4.5\text{ V to }5.5\text{ V}$			$0.3V_{CC}$		$0.3V_{CC}$	V
V_{OH}	HIGH-level output voltage	$V_I = V_{IH}\text{ or }V_{IL}$						
		$I_O = -100\text{ }\mu\text{A};$ $V_{CC} = 1.65\text{ V to }5.5\text{ V}$	$V_{CC} - 0.1$			$V_{CC} - 0.1$		V
		$I_O = -4\text{ mA}; V_{CC} = 1.65\text{ V}$	1.2			1.05		V
		$I_O = -8\text{ mA}; V_{CC} = 2.3\text{ V}$	1.9			1.7		V
		$I_O = -12\text{ mA}; V_{CC} = 2.7\text{ V}$	2.2			2.1		V
		$I_O = -24\text{ mA}; V_{CC} = 3.0\text{ V}$	2.2			2.0		V
		$I_O = -32\text{ mA}; V_{CC} = 4.5\text{ V}$	3.8			3.4		V
V_{OL}	LOW-level output voltage	$V_I = V_{IH}\text{ or }V_{IL}$						
		$I_O = 100\text{ }\mu\text{A};$ $V_{CC} = 1.65\text{ V to }5.5\text{ V}$			0.1		0.1	V
		$I_O = 4\text{ mA}; V_{CC} = 1.65\text{ V}$			0.45		0.65	V
		$I_O = 8\text{ mA}; V_{CC} = 2.3\text{ V}$			0.6		0.8	V
		$I_O = 12\text{ mA}; V_{CC} = 2.7\text{ V}$			0.4		0.6	V
		$I_O = 24\text{ mA}; V_{CC} = 3.0\text{ V}$			0.55		0.8	
		$I_O = 32\text{ mA}; V_{CC} = 4.5\text{ V}$			0.55		0.8	V
I_I	input leakage current	$V_I = V_{CC}\text{ or GND};$ $V_{CC} = 5.5\text{ V}$		± 0.1	± 2		± 10	μA

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Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
I_{OZ}	OFF-state output current	$V_I = V_{IH}$ or V_{IL} ; $V_{CC} = 5.5$ V ; $V_O = V_{CC}$ or GND		± 0.1	± 5		± 10	μA
I_{OFF}	power-off leakage current	V_I or $V_O = 5.5$ V ; $V_{CC} = 0$ V		± 1.6	± 10		± 20	μA
I_{CC}	supply current	$V_I = V_{IH}$ or GND ; $I_O = 0$ A ; $V_{CC} = 5.5$ V		1	10		20	μA
ΔI_{CC}	additional supply current	per input pin ; $V_I = V_{CC} - 0.6$ V ; $I_O = 0$ A ; $V_{CC} = 2.7$ V to 5.5 V		0.3	500		5000	μA
C_i	input capacitance	$\overline{OE}1, \overline{OE}2$ input		3.8				pF
		An input		7.5				pF

[1] All typical values are measured at $V_{CC} = 3.3$ V (unless stated otherwise) and $T_{amb} = 25$ °C.

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10. Dynamic Characteristics

Table 7. Dynamic characteristics

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 6. Typical values measured at $T_{amb} = 25^{\circ}\text{C}$ (unless otherwise noted).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
t_{pd}	propagation delay	An to Yn; see Fig. 4 [2]						
		$V_{CC} = 1.2\text{V}$		65				ns
		$V_{CC} = 1.65\text{ V to }1.95\text{ V}$	3.0	18.5	33.2	3.0	33.6	ns
		$V_{CC} = 2.3\text{ V to }2.7\text{ V}$	2.0	10.0	14.7	2.0	16.8	ns
		$V_{CC} = 3.0\text{ V to }3.6\text{ V}$	1.5	7.0	9.7	1.5	11.3	ns
		$V_{CC} = 4.5\text{ V to }5.5\text{ V}$	1.0	4.8	6.5	1.0	7.6	ns
t_{en}	enable time	$\overline{OE}n$ to Yn; see Fig. 5 [2]						
		$V_{CC} = 1.2\text{V}$		135				ns
		$V_{CC} = 1.65\text{ V to }1.95\text{ V}$	3.5	36.4	68.0	3.5	69.0	ns
		$V_{CC} = 2.3\text{ V to }2.7\text{ V}$	2.5	19.3	31.4	2.5	33.3	ns
		$V_{CC} = 3.0\text{ V to }3.6\text{ V}$	2.0	13.7	21.1	2.0	22.7	ns
		$V_{CC} = 4.5\text{ V to }5.5\text{ V}$	1.5	9.5	13.9	1.5	15.0	ns
t_{dis}	disable time	$\overline{OE}n$ to Yn; see Fig. 5 [2]						
		$V_{CC} = 1.2\text{V}$		136				ns
		$V_{CC} = 1.65\text{ V to }1.95\text{ V}$	3.5	37.2	67.5	3.5	67.9	ns
		$V_{CC} = 2.3\text{ V to }2.7\text{ V}$	2.5	18.9	30.6	2.5	32.1	ns
		$V_{CC} = 3.0\text{ V to }3.6\text{ V}$	2.0	13.5	20.7	2.0	22.0	ns
		$V_{CC} = 4.5\text{ V to }5.5\text{ V}$	1.5	8.7	13.3	1.5	14.1	ns
C_{PD}	power dissipation capacitance	Per buffer ; $V_{CC} = 3.3\text{ V}$ $V_I = \text{GND to }V_{CC}$ [3]		18				pF

[1] Typical values are measured at $T_{amb} = 25^{\circ}\text{C}$ and $V_{CC} = 1.8\text{ V}, 2.5\text{ V}, 3.3\text{ V},$ and 5 V respectively.

[2] t_{pd} is the same as t_{PLH} and t_{PHL} . t_{en} is the same as t_{PZL} and t_{PZH} . t_{dis} is the same as t_{PLZ} and t_{PHZ} .

[3] C_{PD} is used to determine the dynamic power dissipation (P_D in μW).

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \sum (C_L \times V_{CC}^2 \times f_o)$ where:

f_i = input frequency in MHz; f_o = output frequency in MHz

C_L = output load capacitance in pF

V_{CC} = supply voltage in Volts

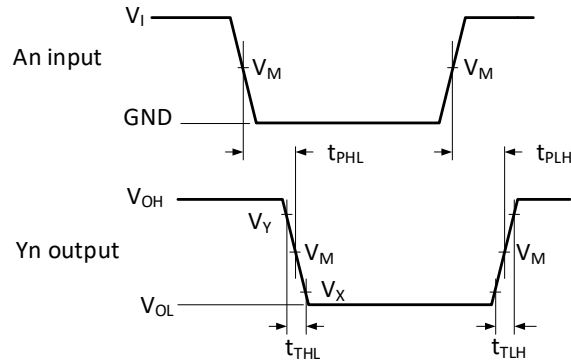
N = number of inputs switching

$\sum (C_L \times V_{CC}^2 \times f_o)$ = sum of the outputs.

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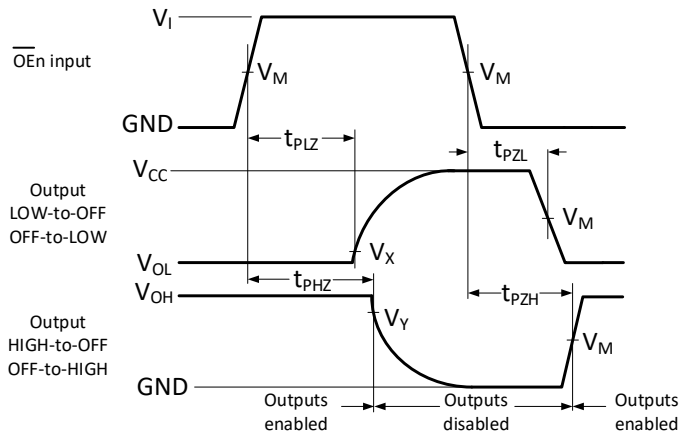
10.1. Waveforms and test circuit



Measurement points are given in Table 8.

V_{OL} and V_{OH} are typical output voltage levels that occur with the output load.

Fig. 4. The input A_n to output Y_n propagation delays



Measurement points are given in Table 8.

V_{OL} and V_{OH} are typical output voltage levels that occur with the output load.

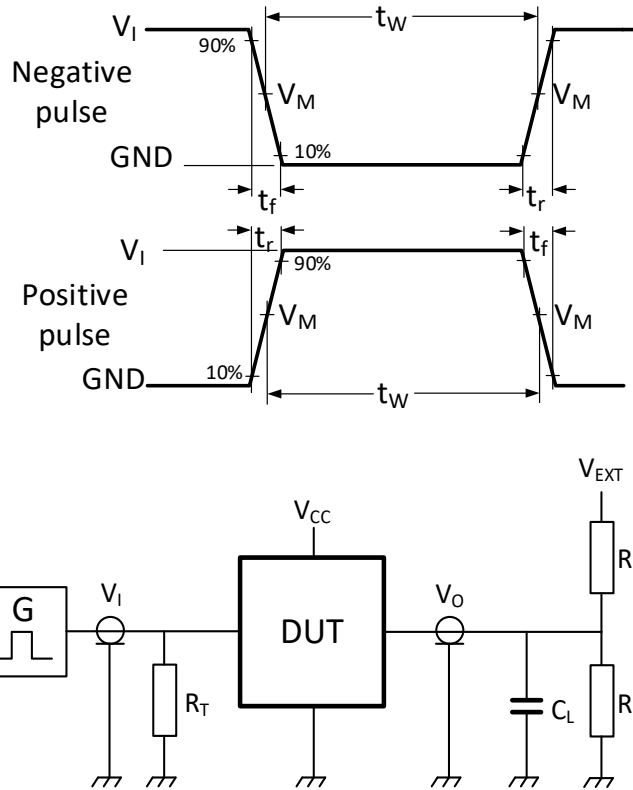
Fig. 5. 3-state enable and disable times

Table 8. Measurement points

Supply voltage	Input		Output		
V_{CC}	V_I	V_M	V_M	V_X	V_Y
1.65 V to 1.95 V	V_{CC}	$0.5V_{CC}$	$0.5V_{CC}$	$V_{OL} + 0.15 \text{ V}$	$V_{OH} - 0.15 \text{ V}$
2.3 V to 2.7 V	V_{CC}	$0.5V_{CC}$	$0.5V_{CC}$	$V_{OL} + 0.15 \text{ V}$	$V_{OH} - 0.15 \text{ V}$
3.0 V to 3.6 V	V_{CC}	$0.5V_{CC}$	$0.5V_{CC}$	$V_{OL} + 0.3 \text{ V}$	$V_{OH} - 0.3 \text{ V}$
4.5 V to 5.5 V	V_{CC}	$0.5V_{CC}$	$0.5V_{CC}$	$V_{OL} + 0.3 \text{ V}$	$V_{OH} - 0.3 \text{ V}$

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Test data is given in Table 9.

Definitions for test circuit:

R_L = Load resistance.

C_L = Load capacitance including jig and probe capacitance.

R_T = Termination resistance should be equal to the output impedance Z_o of the pulse generator.

V_{EXT} = External voltage for measuring switching times.

Fig. 6. Test circuit for measuring switching times

Table 9. Test data

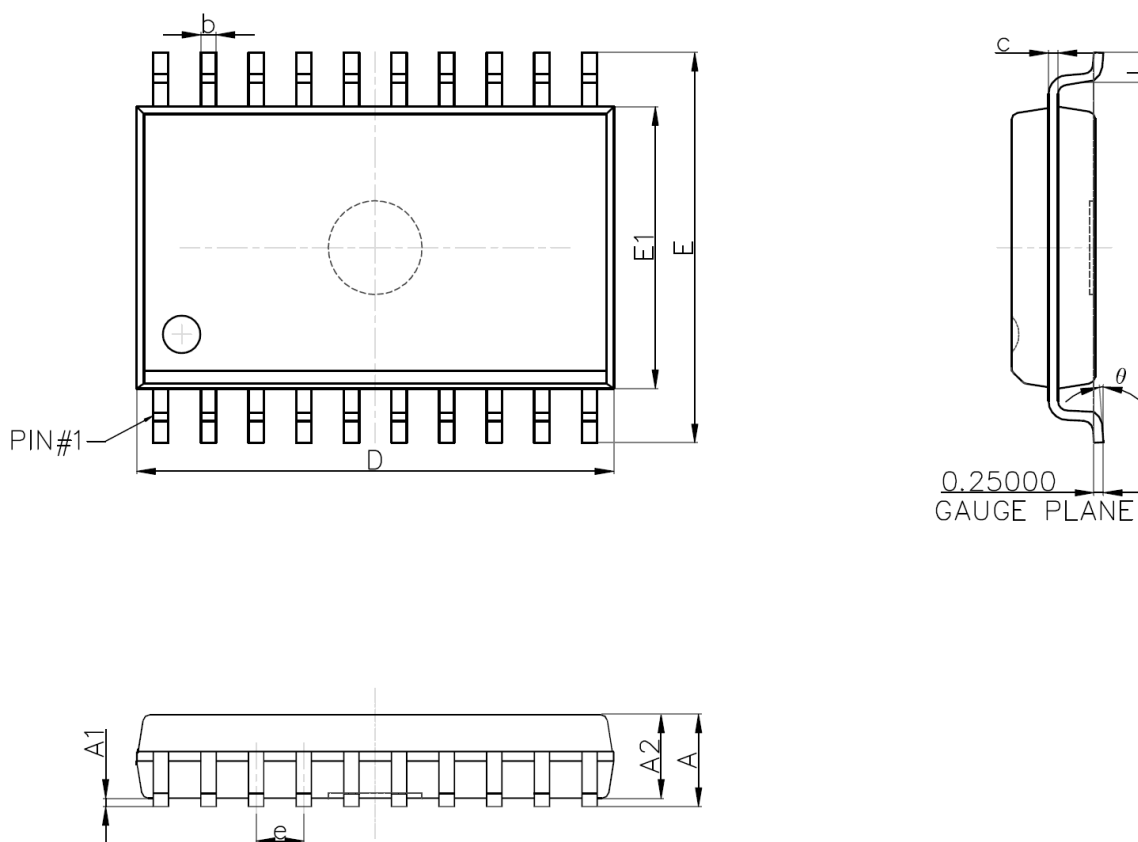
Supply voltage	Input		Load		V_{EXT}		
	V_I	$t_r = t_f$	C_L	R_L	t_{PLH}, t_{PHL}	t_{PZH}, t_{PHZ}	t_{PZL}, t_{PLZ}
1.65V to 1.95V	V_{CC}	≤ 2 ns	15 pF	1 k Ω	open	GND	$2V_{CC}$
2.3 V to 2.7 V	V_{CC}	≤ 2 ns	15 pF	1 k Ω	open	GND	$2V_{CC}$
3.0 V to 3.6 V	V_{CC}	≤ 2 ns	15 pF	1 k Ω	open	GND	$2V_{CC}$
4.5 V to 5.5 V	V_{CC}	≤ 2 ns	15 pF	1 k Ω	open	GND	$2V_{CC}$

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11. Package Outline

SOP-20L

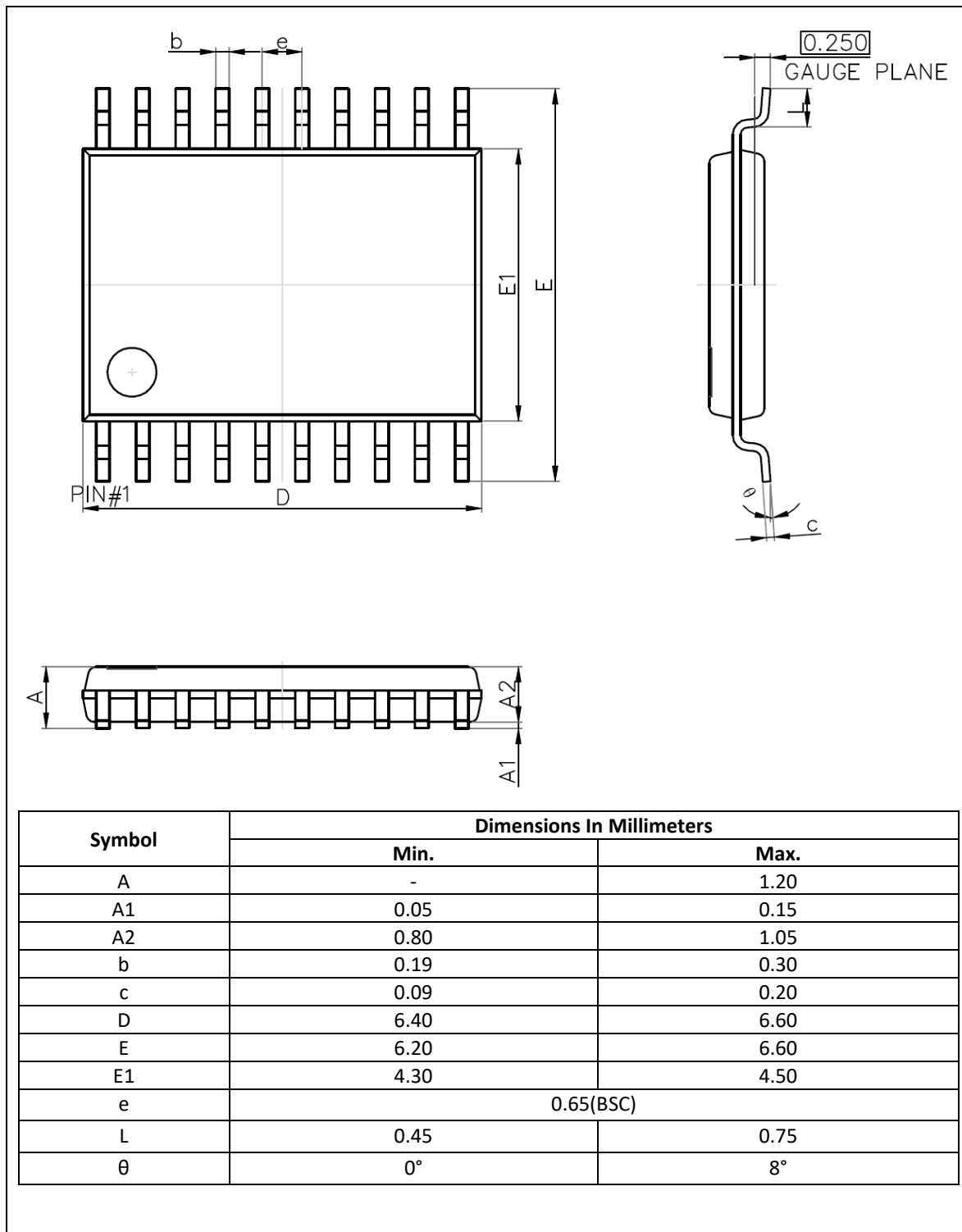


Symbol	Dimensions In Millimeters	
	Min.	Max.
A	2.35	2.65
A1	0.10	0.30
A2	2.10	2.50
b	0.33	0.51
c	0.20	0.33
D	12.50	13.00
E	10.10	10.60
E1	7.40	7.60
e	1.27(BSC)	
L	0.40	1.27
θ	0°	8°

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TSSOP-20L



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12. Tape and Reel Information

12.1. Carrier tape dimensions

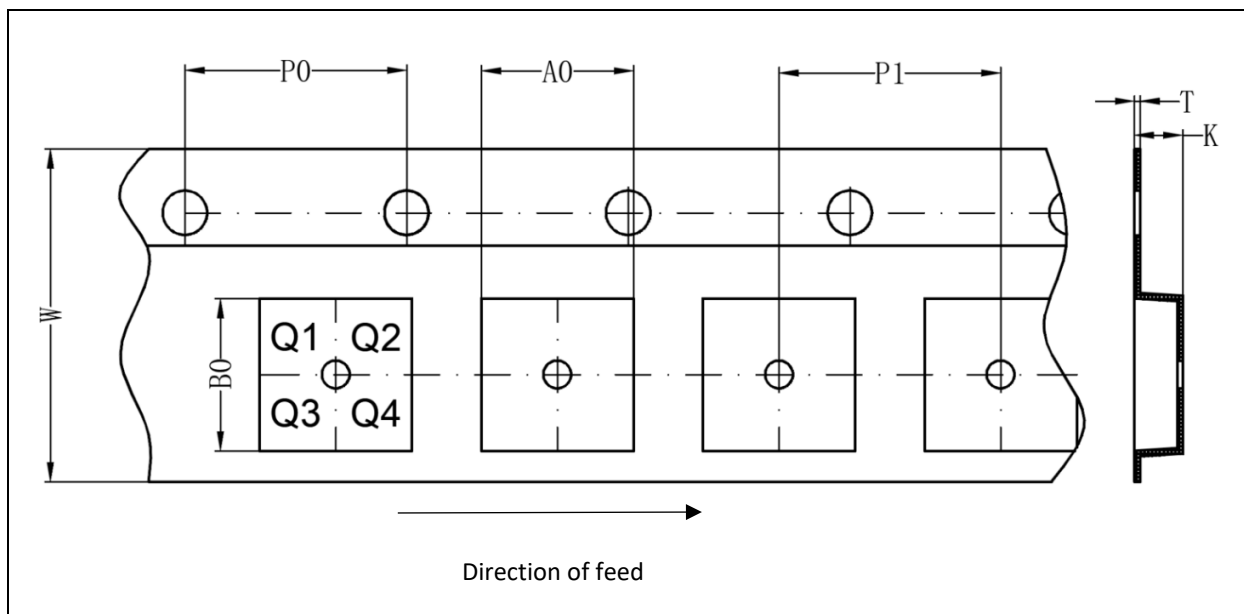


Table 10. Carrier tape dimensions

Package version	A0(mm)	B0(mm)	K0(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
SOP-20L	10.65	13.2	3.2	0.3	12	24	4	Q1
TSSOP-20L	6.85	6.85	1.7	0.22	8	12	4	Q1

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12.2. Reel and box dimensions

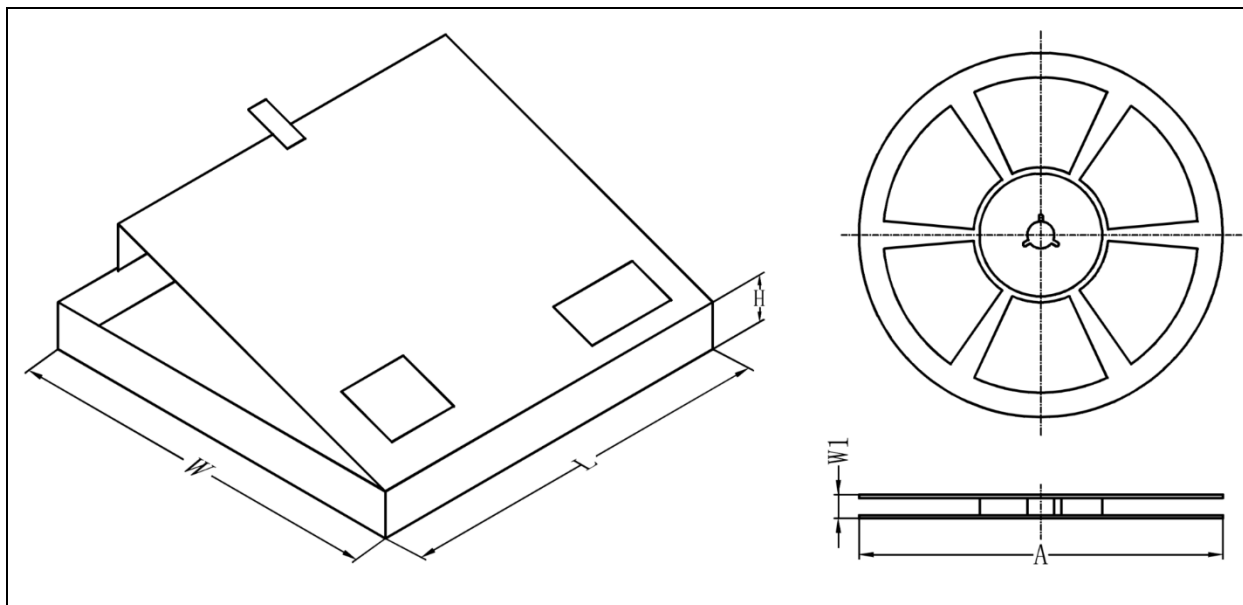


Table 11. Dimensions and quantities

Package version	Type NO. ending	Reel Dimension A (mm)	Max Reel Width W1 (mm)	SPQ (pcs) [1]	Reels per box	Outer box dimensions L×W×H(mm) [2]
SOP-20L	D	330	30.4	2000	1	358x340x50
TSSOP-20L	PW	330	18.4	3000	1	358x340x50

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

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13. Abbreviations

Table 12. Abbreviations

Acronym	Description
CMOS	Complementary Metal-Oxide Semiconductor
DUT	Device Under Test
ESD	ElectroStatic Discharge
HBM	Human Body Model
CDM	Charged Device Model

14. Revision History

Table 13. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EM74LVC541A Rev. 1.0	Aug 28, 2025	Product datasheet		